Chapter 15. The Probabilistic Method

Who Knows What It Looks Like, But It Exists

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15.1 The notion of probability

Motivation. Finite probability is counting over counting; the union bound drives the next section.

Four coin tosses: 5 of 16 outcomes have at least three heads, probability 5/16.

Definition 15.1 (Probability)

Let Ω be a finite set of equally likely outcomes (the **sample space**) and $A \subseteq \Omega$ an **event**. The **probability** of A is $P(A) = \frac{|A|}{|\Omega|}$.

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (sieve); the **complement** $\bar{A} = \Omega - A$ has $P(\bar{A}) = 1 - P(A)$.

Proposition 15.2 (Union bound)

For events $A_1, \ldots, A_n$, $P(A_1 \cup \cdots \cup A_n) \leq P(A_1) + \cdots + P(A_n)$.

Proof. $|A_1 \cup \cdots \cup A_n| \leq |A_1| + \cdots + |A_n|$, since the right side counts each element at least once. ■

Example 15.3

Six numbers are drawn from [36]. A ticket of six numbers contains at least one winning number with probability

$$1 - \binom{30}{6} \Big/ \binom{36}{6} \approx 1 - 0.3048 = 0.6952,$$

not $\frac{6}{36}$.

Example 15.4 (Birthday paradox)

Among 23 people (nobody born on February 29), two share a birthday with probability more than $\frac{1}{2}$, since

$$\frac{365 \cdot 364 \cdots 343}{365^{23}} < \frac{1}{2}.$$

15.2 Non-constructive proofs

Motivation. If a random object is bad with probability < 1, a good one exists.

Theorem 15.5 (Erdős)

$R(k,k) > 2^{k/2}$ for $k \geq 3$.

Proof. Let $n \leq 2^{k/2}$ and colour each edge of K_n by an independent fair coin. A fixed K_k is monochromatic with probability $2^{1-\binom{k}{2}}$; let A_S be this event for S. By Proposition 15.2,

$$P\Big(\bigcup_S A_S\Big) \leq \sum_S P(A_S) = \binom{n}{k} 2^{1-\binom{k}{2}}. \tag{15.1}$$

With $\binom{n}{k} < \frac{n^k}{k!}$ and $n^k \le 2^{k^2/2}$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} < \frac{2^{k^2/2}}{k!} \cdot 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2 \cdot 2^{k/2}}{k!} < 1$$

for $k \ge 3$ (at $k = 3$ it is $\frac{2 \cdot 2^{3/2}}{6}$; $k \mapsto k+1$ multiplies by $\frac{\sqrt{2}}{k+1} < 1$). So some colouring has no monochromatic K_k. ■

With Corollary 13.8: $(\sqrt{2})^k < R(k,k) \le 4^{k-1}$, essentially the best known bounds apart from $(4-\epsilon)^k$.

Theorem 15.6

Let $n, m \ge 2$ with $m \ge 2 \log_2 n$. Then the edges of $K_{n,n}$ can be 2-coloured with no monochromatic $K_{m,m}$.

Proof. Colour randomly: each of the $\binom{n}{m}^2$ copies of $K_{m,m}$ is monochromatic with probability $2 \cdot 2^{-m^2}$. It suffices that $2\binom{n}{m}^2 < 2^{m^2}$: $\binom{n}{m} \le \frac{n^m}{m!}$, $m! \ge 2$ give $2\binom{n}{m}^2 < n^{2m}$, and $n \le 2^{m/2}$ gives $n^{2m} \le 2^{m^2}$. ■

Equivalently, some $n \times n$ 0–1 matrix has no constant $m \times m$ submatrix, $m \approx 2 \log_2 n$; explicit constructions only reach $m = c\sqrt{n}$.

15.3 Independent events

Motivation. Independence: learning B does not change the odds of A.

15.3.1 The notion of independence and Bayes' theorem

For x uniform on $[12]$, divisibility by 2 and by 3 satisfy the product rule; by 2 and by 4 do not.

Definition 15.7 (Independence)

Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$, and **dependent** otherwise.

Definition 15.8 (Conditional probability)

For $P(B) > 0$, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ is the probability of A given B.

Proposition 15.9

For $P(B) > 0$, A and B are independent iff $P(A \mid B) = P(A)$.

Proof. Divide $P(A \cap B) = P(A)P(B)$ by $P(B)$, or multiply back. ■

Example 15.10

Four coin tosses, at least two heads (B). The probability of four heads is $P(A \mid B) = \frac{P(A)}{P(B)} = \frac{1/16}{11/16} = \frac{1}{11}$ (with $P(B) = \frac{1+4+6}{16}$).

Example 15.11

For a random n-permutation, $A = \{p_1 > p_2\}$, $B = \{p_2 > p_3\}$: $P(A) = P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{6}$, so $P(A \mid B) = \frac{1}{3} \neq P(A)$: dependent.

Example 15.12 (Simpson's paradox)

Both colleges admit domestic applicants at a higher rate, the university does not:

	Liberal Arts	Engineering	University
Domestic	10/120 (8.3%)	10/10 (100%)	20/130 (15.4%)
International	1/15 (6.7%)	90/100 (90%)	91/115 (79.1%)

87% of international but 8% of domestic applicants chose the easy Engineering.

Theorem 15.13 (Bayes' theorem, law of total probability)

If A, B are mutually exclusive with $A \cup B = \Omega$ and C is any event, then

$$P(C) = P(C \mid A)P(A) + P(C \mid B)P(B). \tag{15.2}$$

Proof. $P(C) = P(C \cap A) + P(C \cap B)$ and $P(C \cap A) = P(C \mid A)P(A)$, likewise for B. ■

$P(C)$ is a weighted average; in Example 15.12 the weights differ between the groups.

Example 15.14

Exactly one of Andrew (A) and Brandi (B) is guilty; the culprit has a blood type T shared by 5%. If Andrew has type T (X),

$$P(A \mid X) = \frac{P(A \cap X)}{P(X)} = \frac{P(X \mid A)P(A)}{P(X)}. \tag{15.3}$$

$P(X \mid A) = 1$, $P(X \mid B) = \frac{1}{20}$, $P(A) = P(B) = \frac{1}{2}$, so $P(X) = \frac{21}{40}$ by (15.2) and $P(A \mid X) = \frac{20}{21}$.

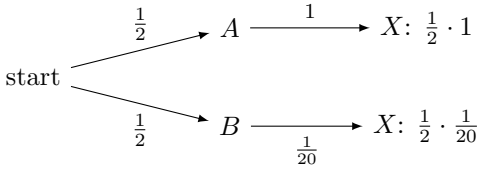


Figure. *Example 15.14 as a tree:* $P(A \mid X) = \frac{1/2}{1/2+1/40} = \frac{20}{21}$.

15.3.2 More than two events

Pairwise independence is not enough: for $x \in [10]$, $y \in [20]$, $A = \{x \text{ odd}\}$, $B = \{y \text{ odd}\}$, $C = \{x - y \text{ odd}\}$ are pairwise independent with probability $\frac{1}{2}$, but $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$.

Definition 15.15 (Mutual independence)

$A_1, \dots, A_n$ are **independent** if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every nonempty $\{i_1, \dots, i_k\} \subseteq [n]$.

Theorem 15.16 (Bayes' theorem, general version)

If $A_1, \dots, A_n$ are pairwise disjoint with union Ω and C is any event, then $P(C) = \sum_{i=1}^{n} P(C \mid A_i)P(A_i)$ (terms with $P(A_i) = 0$ omitted).

Proof. $C = \bigsqcup_i (C \cap A_i)$ and $P(C \cap A_i) = P(C \mid A_i)P(A_i)$. ∎

Monty Hall. Door 1 is picked; the host opens door 2 showing a goat (B). With $A_i =$ "car behind door i",

$$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{\sum_i P(B \mid A_i)P(A_i)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \frac{1}{3}, \tag{15.4}$$

so switching wins with probability $\frac{2}{3}$.

15.4 Expected values

Motivation. Linearity splits complicated counts into indicators; some outcome beats the average.

A **random variable** is a function $X : \Omega \to \mathbb{R}$.

Definition 15.17 (Expectation)

If X takes finitely many values S, its **expected value** is $E(X) = \sum_{i \in S} i\, P(X = i)$, where $P(X = i) = \frac{|\{u : X(u) = i\}|}{|\Omega|}$. Equivalently,

$$E(X) = \frac{1}{|\Omega|} \sum_{u \in \Omega} X(u). \tag{15.5}$$

(Write E_Ω if needed; countable ranges need convergent sums.)

Definition 15.18 (Independent variables)

X, Y are **independent** if $P(X = s, Y = t) = P(X = s)P(Y = t)$ for all s, t.

15.4.1 Linearity of expectation

Theorem 15.19 (Linearity)

(1) $E(X + Y) = E(X) + E(Y)$ for random variables on the same space.
(2) $E(cX) = cE(X)$ for $c \in \mathbb{R}$.

Proof.

Step 1 (equally likely outcomes). By (15.5): $\sum_u (X + Y)(u) = \sum_u X(u) + \sum_u Y(u)$, $\sum_u cX(u) = c \sum_u X(u)$.

Step 2 (without uniformity). With values x_i, y_j,

$$E(X+Y) = \sum_{i,j}(x_i+y_j)P(X=x_i, Y=y_j) = \sum_i x_i P(X=x_i) + \sum_j y_j P(Y=y_j);$$

for (2), $P(cX = cx_i) = P(X = x_i)$ if $c \neq 0$. ■

No independence is needed. An **indicator variable** of an event has expectation its probability.

i is a **valley** of p if $2 \le i \le n-1$ and $p_i < p_{i-1}$, $p_i < p_{i+1}$.

Theorem 15.20

For $n \ge 2$, a random n-permutation has on average $\frac{n-2}{3}$ valleys.

Proof. The indicator Y_i of a valley at i has $E(Y_i) = \frac{1}{3}$ (each of p_{i-1}, p_i, p_{i+1} equally likely smallest), so $E(\sum_{i=2}^{n-1} Y_i) = \frac{n-2}{3}$. ■

Theorem 15.21

A random n-permutation has on average one fixed point.

Proof. The indicator X_i of $p_i = i$ has $E(X_i) = \frac{1}{n}$, so

$$E(X) = \sum_{i=1}^{n} E(X_i) = n \cdot \frac{1}{n} = 1. \tag{15.6}$$

■

15.4.2 Existence proofs using expectation

Theorem 15.22

If X takes finitely many values with maximum j, then $j \ge E(X)$. Hence some outcome u has $X(u) \ge E(X)$.

Proof. $E(X) = \sum_{i \in S} i\, P(X=i) \le j \sum_{i \in S} P(X=i) = j$. ■

Theorem 15.23

A simple graph G on $[n]$ ($n \ge 2$) with $m \ge 1$ edges has a bipartite subgraph with more than $m/2$ edges.

Proof. Let Ω be the $2^{n-1}-1$ splits $\{A, B\}$ of $[n]$ into nonempty parts and X the number of crossing edges. Edge uv crosses in 2^{n-2} splits (place the other $n-2$ vertices freely), so

$$E(X_i) = P(X_i = 1) = \frac{2^{n-2}}{2^{n-1}-1} > \frac{1}{2}.$$

Then $E(X) > \frac{m}{2}$; apply Theorem 15.22. ■

Example 15.24 (Betweenness)

Given triples $L_i = (a_i, b_i, c_i)$ of distinct elements of $[n]$, $i \le k$, say that p satisfies L_i if b_i lies between a_i and c_i in p. Some n-permutation satisfies at least $k/3$ of the triples.
Solution. For random p, L_i holds with probability $\frac{1}{3}$; the expected number satisfied is $\frac{k}{3}$ (Theorem 15.22).

15.4.3 Conditional expectation

$E(X \mid A) = \sum_i i\, P(X = i \mid A)$ for $P(A) > 0$.

Theorem 15.25

If $A_1, \dots, A_n$ are pairwise disjoint with union Ω (and positive probabilities), then $E(X) = \sum_{i=1}^{n} E(X \mid A_i) P(A_i)$.

Proof. Theorem 15.16 with $C = \{X = j\}$, times j, summed over j:

$$E(X) = \sum_j j \sum_i P(X = j \mid A_i) P(A_i) = \sum_i P(A_i) \sum_j j\, P(X = j \mid A_i) = \sum_i E(X \mid A_i) P(A_i).$$

■

Example 15.26

Three die throws, the first ≥ 4. It is even (A_1) with probability $\frac{2}{3}$, giving on average $1 + 2 \cdot \frac{1}{2} = 2$ even throws, and odd (A_2) with $\frac{1}{3}$, giving 1. Expected even throws: $\frac{2}{3} \cdot 2 + \frac{1}{3} \cdot 1 = \frac{5}{3}$.

Example 15.27

A team wins each game independently with probability $\frac{3}{4}$. Given at least three wins in the first four games (B), what is the expected number of wins in 12 games?
Solution. A_1, A_2: exactly 3 / all 4 first games won. $P(A_1) = 4 \cdot \frac{1}{4}(\frac{3}{4})^3 = \frac{27}{64}$, $P(A_2) = (\frac{3}{4})^4 = \frac{81}{256}$, so $P_B(A_1) = \frac{4}{7}$, $P_B(A_2) = \frac{3}{7}$; the last eight add 6 on average. By Theorem 15.25 in B,

$$E_B(X) = 9 \cdot \frac{4}{7} + 10 \cdot \frac{3}{7} = \frac{66}{7},$$

more than the unconditional 9.

15.5 Selected exercises

Motivation. Indicators for an existence proof and for a second moment.

Exercise 15.17 (Tournaments with many Hamiltonian paths)

Prove that for every $n \ge 1$ there is a tournament on n vertices with at least $\frac{n!}{2^{n-1}}$ Hamiltonian paths. What can be said about Hamiltonian cycles?

Solution.

Step 1 (random tournament). Orient each edge of K_n independently with probability $\frac{1}{2}$ each way.

Step 2 (indicators). For a permutation σ, the indicator X_σ of the path $\sigma_1 \to \cdots \to \sigma_n$ fixes $n-1$ edge directions: $E(X_\sigma) = 2^{-(n-1)}$.

Step 3 (expectation). $E(\sum_\sigma X_\sigma) = \frac{n!}{2^{n-1}}$ (Theorem 15.19); apply Theorem 15.22.

Cycles ($n \geq 3$): $(n-1)!$ cyclic orders (Example 6.10), each fixing n directions, give a tournament with $\geq \frac{(n-1)!}{2^n}$ Hamiltonian cycles. ■

Exercise 15.18 (The variance of the number of fixed points)

For a random variable Y let $\mathrm{Var}(Y) = E((Y - E(Y))^2)$.
(a) Prove that $\mathrm{Var}(Y) = E(Y^2) - E(Y)^2$.
(b) Let $X(p)$ be the number of fixed points of a random n-permutation p, $n \geq 2$. Prove that $\mathrm{Var}(X) = 1$.

Solution.

(a) With $\mu = E(Y)$, by Theorem 15.19,

$$E((Y-\mu)^2) = E(Y^2) - 2\mu E(Y) + \mu^2 = E(Y^2) - \mu^2.$$

(b)

Step 1 (indicators). $X = \sum_i X_i$, X_i the indicator of $p(i) = i$, $E(X_i) = \frac{1}{n}$, $E(X) = 1$.

Step 2 (second moment). As $X_i^2 = X_i$,

$$X^2 = \sum_i X_i + \sum_{i \neq j} X_i X_j,$$

and $E(X_i X_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$ for $i \neq j$, so

$$E(X^2) = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 2.$$

Step 3 (conclusion). $\mathrm{Var}(X) = 2 - 1^2 = 1$ (like Poisson(1); for $n = 1$ it is 0). ■